

Assessing Whisper automatic speech recognition and *WER* scoring for elicited imitation: Steps toward automation

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ARTICLE INFO

Keywords:

Elicited imitation
Oral proficiency assessment
Automatic speech recognition
Computer automation

ABSTRACT

Elicited imitation (EI) has received considerable attention in the field of SLA as a psycholinguistic method for oral proficiency assessment. However, EI tests are perhaps underutilized because of the need for time-consuming scoring by hand. A clear solution to this is computer automation, which requires two key components to work in parallel: (1) an accurate automatic speech recognition (ASR) system to transcribe EI response speech into text, and (2) a computational scoring metric to assess the resulting text. In this paper, we test the feasibility of automating EI scoring with the open-source Whisper ASR system and the Word Error Rate (*WER*) scoring metric which measures deviation from the original prompt. 30 Japanese L1 participants took a 30-item test, resulting in 900 English L2 learner EI sentence responses which were transcribed by the two authors and Whisper ASR. The intraclass correlation coefficient (ICC) between the error rates of the human raters and Whisper across all items was $ICC = 0.929$, 95 % CI [0.921, 0.936] indicating very strong alignment. We then compared automated test scores using *WER* to those done manually by human raters using a traditional ordinal-scale scoring method developed by Ortega et al. (2002) and found a robust correlation of $r = 0.969$, 95 % CI [0.935, 0.985] across overall participant scores. These findings show that the combination of the Whisper ASR system and the *WER* scoring metric result in EI test scores that align almost perfectly with and are arguably an improvement on currently accepted methods.

Background

Elicited imitation (EI) has received considerable attention in the field of SLA as a psycholinguistic method for oral proficiency assessment. In an EI task, a subject hears a sentence in the target language (elicitation) and, after a brief pause, attempts to repeat it as accurately as possible (imitation). This may sound simple, but in theory, when a sentence is long enough to exceed the working memory capacity—approximately eight to ten syllables—it requires quick processing and reconstruction (Underhill, 1987). An example EI prompt we have used in this study is “many people waited to enter the park that morning.” This sentence has 14 syllables and features a past tense verb, an infinitive phrase, an adverbial phrase, and three noun phrases. Because of this sentence’s length and complexity, successful performance indicates advanced language processing competence and the ability to quickly comprehend and reconstruct complex linguistic information (Van Moere, 2012). The real-time demands of EI show the extent to which a learner has internalized the features and structure of the language.

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Elicited imitation has been used in second language research since the 1970s (Hamayan et al., 1977; Naiman, 1974) as a measure of L2 comprehension and production skills. EI methods were further refined in the 1980s (Gallimore & Tharp, 1981), and efforts were made to understand what linguistic features contribute to EI item difficulty (Perkins et al., 1986). Interest in EI as a measure of global oral proficiency took off in the early 2000s following the creation of a series of parallel EI tests in English, Spanish, German, and Japanese as well as a detailed ordinal-scale scoring rubric by Ortega et al. in 2002 (hereafter referred to as the Ortega test and rubric). This initial study found a strong correlation of $r = 0.88$ between the Spanish version EI test and a Simulated Oral Proficiency Interview (SOPI) with a sample size of $n = 28$. A much larger study ($n = 156$) using a different English EI test found a moderate to strong correlation of $r = 0.658$ with the English Oral Proficiency Interview (OPI) (Graham et al., 2008).

EI task design can be adjusted to target specific constructs, as seen in a study by Erlam (2006) which used EI to measure implicit knowledge of 17 specific grammatical structures such as third person -s and plural -s using both grammatical and ungrammatical prompts. An example ungrammatical prompt from the study is “Everyone loves comic books and read them” (pg. 475). After hearing each statement prompt, participants first had to write on a mark sheet their belief about whether they felt the prompt was true or false before being asked to repeat it out loud in correct English. Native speakers were found to correct ungrammatical prompts 91 % of the time, while L2 learners corrected them only 35 % of the time, indicating less implicit knowledge.

Two large meta-analyses of EI studies have also been conducted. A survey of 76 EI studies by Yan et al. (2016) showed EI could effectively measure global language proficiency and distinguish speakers of different proficiency levels (Hedges’ $g = 1.34$). Another more recent meta-analysis of 46 studies (Kostromitina & Plonsky, 2022) found a strong overall correlation of $r = 0.74$ between EI test scores and standardized proficiency tests such as IELTS and TOEFL.

Critics of EI might claim that participants are memorizing or parroting prompts rather than reconstructing them. However, an early study by Hamayan et al. (1977) used a number of ungrammatical prompts in an EI test and found that many of the participants corrected the mistakes without being asked. Had they just been parroting, they would have repeated the mistakes. Similarly, Gathercole and Baddeley (1993) used EI prompts which included non-words and found that subjects had more difficulty repeating sentences with less wordlike non-words. Again, this would not be the case if it were just parroting. Several studies have also investigated the role of working memory in EI, finding only weak and non-significant correlations between working memory measures and EI performance (Kim et al., 2016; Norris et al., 2022; Okura & Lonsdale, 2012; Park et al., 2020) and all finding language-related measures to be the strongest predictors.

An EI-based oral proficiency test generally involves a number of prompts of increasing length within a specified syllable range. The most widely known EI test is the previously mentioned Ortega test which comprises 30 prompts ranging from 7 to 19 syllables in length. Test responses are recorded and then manually scored on a 5-point scale by a human judge, giving a total combined score used to measure oral proficiency. The Ortega test has been validated by showing correlations with other proficiency tests (Kostromitina & Plonsky, 2022) in several studies, translated into various languages, and is freely available. Because of this, it remains the standard used in many EI studies, as evidenced by work published in a special issue of *Research Methods in Applied Linguistics* (Solon & Park, 2024).

Automating elicited imitation tests

The biggest scaling obstacle for any kind of oral proficiency test evaluation is the need for trained judges. This applies to elicited imitation tests such as the Ortega test as well; EI tests are perhaps underutilized because the need for manual scoring makes them time consuming. A clear solution to this is computer automation, which we explore in this paper. Automating oral proficiency assessment requires two key components to work in parallel: (1) an accurate automatic speech recognition (ASR) system to transcribe speech into text, and (2) computational scoring metrics to assess features of the resulting text.

ASR systems are primarily designed to recognize clear native speech. Their accuracy is significantly lower when dealing with non-native or accented speech (Tomokiyo, 2000). This has been the primary bottleneck preventing reliable automated EI until now. From 2008–2012 there was a concerted effort by Daryl Lonsdale and his associates at Brigham Young University (BYU) to automate EI with ASR (Cox & Davies, 2012; Graham et al., 2008; Lonsdale & Christensen, 2011). The BYU group used the open-source Sphinx ASR System (Lee, 1989), which struggled to provide accurate transcriptions of English L2 learner responses. Considerable amounts of fine-tuning to the ASR parameters and language models were required to reach modest levels of transcription accuracy for non-native speech. Such adjustments included forced alignment (i.e., constraining the language model to only recognize the words in the prompt), and removing words or constructions that are problematic for the ASR. Although the papers produced by this group are intriguing and raise important questions, the work that was produced never led to any real advances in the way EI was conducted.

A later attempt by Ashwell and Elam (2017) had similar accuracy issues using the Google Web Speech ASR system. Their approach achieved overall accuracy of 89.4 % with native English speech but only 65.7 % accuracy with English L2 learner speech.

The Whisper ASR system

In 2023, OpenAI released an open-source transformer-based ASR system called Whisper which was trained on massive multilingual data sets and could transcribe speech with error rates approaching those of transcriptions done by humans (Radford et al., 2022). The Whisper ASR system had four key advantages: (1) it reached closer to human level accuracy than any previous ASR system; (2) it was trained on multi-lingual datasets and could handle non-native and heavily accented speech; (3) it had a high tolerance for background noise, so it could handle recordings of varying quality from smartphones and computers; (4) it did not require any fine-tuning or forced alignment to adjust for these situations (Radford et al., 2022). Whisper has been used in a recent tutorial on EI (Kim et al., 2024), and

we use the Whisper ASR system in the present study. Examples of Whisper ASR's performance in different contexts including recordings of varying quality in different languages and even transcription of song lyrics from music can be seen at <https://replicate.com/openai/whisper/examples>.

Using Whisper

Whisper is free and open-source software which can be downloaded and installed on a local computer with a GPU. However, it is often easier to use a commercial application programming interface (API) which allows you to pay a small fee to run computer code on remote high-end computers and removes the need for your own GPU. OpenAI, the company which created Whisper, offers an API, but there are many other APIs which are cheaper and have additional features. The present study used an API offered by Replicate (<https://replicate.com>), a cloud-based machine learning platform that provides a large library of machine learning models. While using the API requires programming knowledge, Replicate also offers an online graphical user interface for anyone who would like to try it at <https://replicate.com/openai/whisper>. This can be used to transcribe individual audio files one at a time within a limit and can demonstrate the ASR system's capabilities.

Once a Replicate account is created and payment details are added, an API token can be generated in the account settings. This token is essentially a password that allows your code to run on Replicate's computers. By default, the Replicate Whisper API runs on an Nvidia T4 high memory GPU. Each machine learning model has different API settings, but the documentation for the Whisper model API can be found at <https://replicate.com/openai/whisper/api>. We used the API in a script written in the Python programming language which batch processed all 900 of our recordings and took 1030 s (an average of 1.14 s per response).

The cost of running the Whisper API offered by Replicate is USD \$0.000225 per second of processing time, or how long it takes the cloud computers to process your data (not the length of the audio recordings). The 1030 s needed to process all 900 responses from this study cost USD \$0.23, less than one cent for each 30-item test. In comparison, the OpenAI Whisper API charges USD \$0.006 per minute or USD \$0.0001 per second of audio (not processing time like Replicate) rounded to the nearest second. The total length of all our response recordings combined was 1 h 23 m 27 s, or 5007 s. This would cost USD \$0.50 to process all of our responses, more than double the cost of the Replicate API.

The Word Error Rate (WER) metric

Using computers to score speech productions, thus reducing the need for human raters, has also been an ongoing challenge. This is because conventional oral proficiency test questions such as those on the OPI or IELTS are open-ended with no single correct answer. In contrast to this, EI has standardized prompts and responses, resulting in a fixed answer space that can be assessed much more easily (Zhang et al., 2020). This makes EI ideal for computer automation. If accurate ASR transcription is now possible, then all that is required to assess EI tests is to compare the test-taker's response transcription to the original prompt and calculate an accuracy score.

Word Error Rate (WER) is a computational scoring metric that compares two strings of text at the word level and identifies correct words as well as substitutions, insertions, and deletions. It was originally created to test the accuracy of ASR systems, but we believe it is perfectly suited for scoring EI as well. It can be performed nearly instantaneously; our Python script calculated the WER of 900 response transcriptions in 0.036 s on a desktop computer. WER accounts for all word-level differences between the response and the prompt and is thus much more granular than a conventional ordinal-scale method, such as the Ortega scoring rubric, which may assign

```
import jiwer

out = jiwer.process_words(
    ["many people waited to enter the park that morning"],
    ["the many people wait to enter park this morning"],
)

print(jiwer.visualize_alignment(out))

sentence 1
REF: *** many people waited to enter the park that morning
HYP: the many people    wait to enter *** park this morning
      I                S          D      S

number of sentences: 1
substitutions=2 deletions=1 insertions=1 hits=6

mer=40.00%
wil=55.56%
wip=44.44%
wer=44.44%
```

Fig. 1. Demonstration of WER using the jiwer Python package in Jupyter Notebook.

a sentence level score based on a single mistake. Kostromitina and Plonsky's 2022 meta-analysis found the strongest correlation between EI and other proficiency measures in studies that used interval-scale scoring methods (such as *WER*) as opposed to ordinal or binary scales due to higher granularity.

To illustrate how *WER* works, we can look at our earlier example EI prompt: "many people waited to enter the park that morning." If a test-taker responds by saying "the many people wait to enter park this morning," *WER* identifies the insertion of the initial *the*, the substitutions of *wait* for *waited* and *this* for *that*, and the deletion of *the* before *park*, resulting in a *WER* of 0.44 (or 44 %). The output of a simple Python script using the *jiwer* package (Vaessen, 2018 - documentation can be found at <https://jitsi.github.io/jiwer/>) in Jupyter Notebook (Kluyver et al., 2016) is shown in Fig. 1. The *WER* score can also be truncated so that the maximum is 1 (or 100 %), preventing a negative accuracy score in the case of a response which contains many more words than the prompt. Because this is an error rate, the maximum *WER* of 1 would mean the participant would receive an accuracy score of 0 % on their utterance. The *WER* score of 0.44 in the above example means a 56 % accuracy in producing the sentence ($1 - 0.44 = 0.56$).

The 5-point ordinal scale rubric devised by Ortega et al. (2002) may have been necessary for human raters to efficiently score a large number of responses, but a more nuanced and granular score is clearly preferable. With the Ortega scale, sentences are given a score between 0 and 4, with a 0 given for silent or unintelligible responses and a 4 given for exact repetition. The criteria for differentiating between a score of 2 or 3, however, is often decided by a single mistake, and certain grammatical features are prioritized above others due to their impact on the meaning of the sentence. The clearest example of this is a plural mistake—inserting or omitting the final {-s} on a noun. A response with this single mistake is scored as a 2, regardless of other mistakes or the length of the prompt. Meanwhile, a response with different mistakes may receive a 3 if the exact meaning is not altered. In many cases it is difficult to decide if the meaning has changed or not, and the score may depend on the rater's interpretation of the scoring criteria.

Fig. 2 illustrates scoring differences between *WER* and the Ortega method by showing the scores for two slightly different responses (HYP in Fig. 2) to the same prompt (REF in Fig. 2) from our pilot test. The prompt is 18 syllables long and relatively complex. Both responses contain two mistakes—one substitution (S) and one deletion (D)—and therefore receive the same *WER* score of 0.1538 (84.62 % accuracy) based on the length of the original prompt. However, the changes in sentence 1 do not affect the meaning and would receive a 3 on the Ortega scale, while sentence 2 contains a plural mistake and would receive a 2 on the Ortega scale. If similar mistakes occurred in a shorter prompt, *WER* would adjust the value of each word, while the Ortega scores would not change. Under the Ortega system, an 18-syllable response with a single mistake would receive the same score as a 9-syllable response with a single mistake.

In contrast to the Ortega rubric, *WER* identifies all word level differences between the response and the prompt and penalizes each by an equal amount based on the prompt length. These differences can easily be logged, collected, and analyzed to identify trends or patterns within the testing population. While the global meaning of the sentence is not taken into account, we feel it is a more reproducible (i.e., reliable) and logical way to score responses than relying on human judges to consistently follow all scoring criteria. Even given these differences, a recent study (Isbell et al., 2023) compared several commercial ASR systems—not including Whisper—and computational scoring metrics to Ortega scores and found them to highly correlate at $r = 0.91$ – 0.96 depending on the ASR model. This shows there is not that much difference between some computational metrics and the Ortega method in the outcome, but there is a large difference in the amount of time and work needed to reach that outcome. Our present study adds to this evidence by comparing *WER* scoring to Ortega scoring.

Research questions

This study aims to validate the two key components required for fully automating EI tests: automatic speech recognition and computational scoring. The research question are as follows:

```

sentence 1 (D-013-183)
REF: in a tea ceremony people drink tea out of bowls instead of cups
HYP: in a tea ceremony people drink tea from ** bowls instead of cups
      S D

substitutions=1 deletions=1 insertions=0 hits=11

WER = 0.1538
Accuracy = 84.62%
Ortega Score = 3 (75%) / minor changes that don't affect content

sentence 2 (D-022-183)
REF: in a tea ceremony people drink tea out of bowls instead of cups
HYP: in * tea ceremony people drink tea out of bowl instead of cups
      D S

substitutions=1 deletions=1 insertions=0 hits=11

WER = 0.1538
Accuracy = 84.62%
Ortega Score = 2 (50%) / minor changes that affect content

```

Fig. 2. Comparison of *WER* and Ortega scores on two responses.

1. Can Whisper ASR accurately transcribe the utterances of Japanese users of English?

We test this question by measuring how similar the results of Whisper ASR transcriptions of EI responses are to human transcriptions of the same responses in terms of deviation from the original prompts measured by *WER*. This provides reliability evidence for the ASR component of automated EI testing.

2. Can *WER* scoring distinguish accuracy as well as Ortega scoring?

We test this question by comparing the results of an EI test when scored by *WER* and when scored using the Ortega method done by human raters. This provides convergent validity evidence by determining whether the computationally derived *WER* scores align with the established Ortega scoring method, which serves as a criterion measure for oral proficiency assessment in this context.

It is important to note that the *WER* metric is being used in different ways for each of these research questions. For RQ1, *WER* is being used to measure the amount of deviation from the original prompts found in both Whisper ASR transcriptions and human transcriptions. This does not mean that the transcriptions must be identical, especially in the case of unintelligible responses, but rather that the raters should agree on the amount of error. Both human raters and Whisper ASR attempt to transcribe exactly what is being said in the response recording, but there is no ground truth to compare to beyond what can be heard.

For RQ2, the same Whisper ASR *WER* score is being used to score the entire EI test, and this score is being compared to Ortega scoring done by humans. In this case the original prompt is the ground truth.

Materials and methods

Participants

A combined 30 university undergraduates were recruited from the authors' home universities. All were L1 Japanese speakers who had been studying English for at least six years. 14 were male, 16 were female, and their ages ranged from 18 to 45, with an average age of 20.

To include a range of English proficiency levels, participants were recruited from different academic contexts. 22 participants were English majors at the first author's university. For descriptive purposes, we note that these students' TOEFL ITP (Institutional Testing Program, paper-based) scores ranged from 480 to 650 with an average of 572. The remaining 8 participants were primarily non-English majors at the second author's university with TOEIC scores ranging from 755 to 935 with an average of 845.

While these standardized test scores provide context about our sample's general English proficiency, we acknowledge that they do not specifically isolate oral proficiency, which is most relevant to our study. This absence of readily available oral proficiency measurements highlights the specific problem our research seeks to address. Participants were incentivized to participate in the study by providing a 500-yen QUO gift card upon completion of the test.

Materials

The 30 EI sentence prompts used in the test (found in [Appendix A](#)) were extracted from the Simple Wikipedia Corpus ([Leipzig](#)

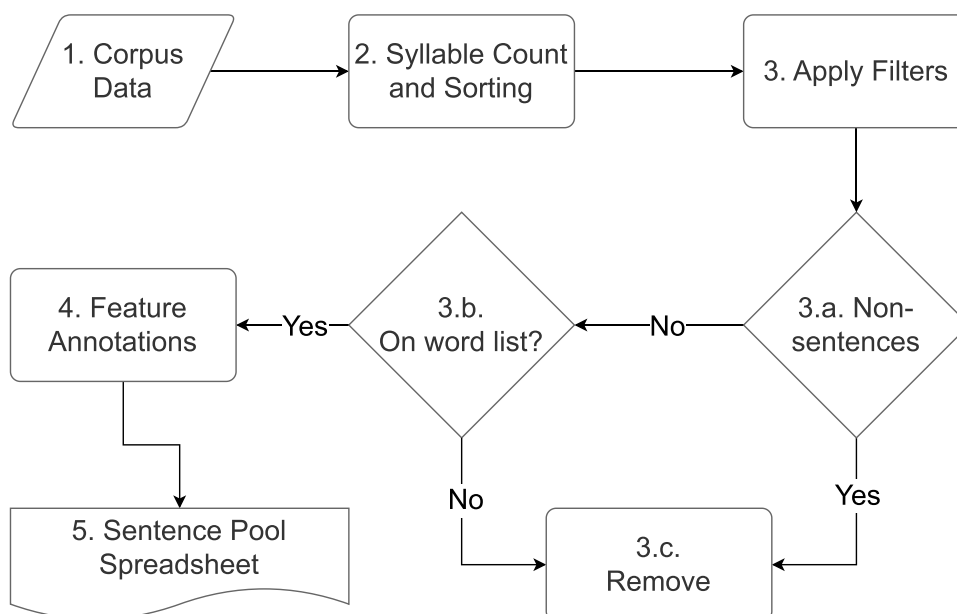


Fig. 3. EI item generation flowchart.

Corpora Collection, 2021) using a Python script. The 5-step process is laid out in Fig. 3: (1) Text files containing corpus data are loaded. (2) The syllable length of each sentence is calculated by checking words in the Carnegie Mellon University Pronouncing Dictionary (CMUdict - <http://www.speech.cs.cmu.edu/cgi-bin/cmudict>) using the NLTK Python package (Bird et al., 2009); sentences that fall within a specified syllable range are extracted and sorted. (3) Two filters are applied to remove (3a) fragments and non-sentences and (3b) sentences that contain low-frequency words - in this case we used the New General Service List (Browne et al., 2013), which is a list of the 2800 most common words in general English, but other lists can be easily used. (4) The resulting sentences are annotated for syntactic features (tree depth, constituent count, and dependency relations, etc.), average word frequency and prevalence, and the presence of 25 grammatical features (such as different verb tenses, subordinate clauses, plurals, modals, demonstratives, etc.) using the NLTK package. (5) The sentences and annotations are output to an item pool spreadsheet according to syllable length.

From this pool, three sentences per syllable count from 9 to 18 syllables were ultimately chosen by the authors, with one sentence deliberately being a shorter, denser sentence and the other two containing a larger number of words. Future test development will be more rigorous in sentence selection, but our intent in this study was to explore the feasibility of automation.

Testing procedure

The EI test was administered in computer classrooms so that participants could be monitored. Each participant took the test online using the audio recording function of a commercial learning management system called ZenGengo. The platform was not designed for EI, but it was adequate for data collection. Participants heard the EI sentence prompts via the website through headphones with an attached microphone. After hearing each sentence one time, participants pushed a button to begin recording their repetition. Although no other activities besides the test were taking place in the classroom, the sound of other participants repeating sentences could sometimes be heard in the background of the recordings. We say this to point out that Whisper did not have any trouble dealing with this kind of background noise. Because of the limitations of the platform we used for this initial test, participants could pause between prompts, but all participants had finished the test within 15 min.

RQ1: Whisper ASR Transcription Assessment Procedure

In order to assess the ability of Whisper ASR to accurately transcribe EI responses, we needed to compare Whisper transcriptions to human transcriptions of the same responses. All 900 EI test response recordings were downloaded from the learning management system to a local computer. Both authors independently listened to all response recordings and transcribed them without reference to the original prompts. Then, a Python script was written to transcribe the recordings using Whisper ASR through the Replicate API.

Next, the WER for each response transcription was calculated by comparing it to the original prompt using the jiwer package

Table 1

Average WER for each transcriber over 30 items for 30 participants.

Participant	Rater 1 Average Transcription WER	Rater 2 Average Transcription WER	Whisper Average Transcription WER
D-001	0.1012	0.1151	0.1038
D-002	0.1445	0.1522	0.1287
D-003	0.2546	0.2808	0.2522
D-004	0.1590	0.1530	0.1252
D-006	0.3066	0.3203	0.3066
D-007	0.2885	0.2888	0.2814
D-008	0.1314	0.1334	0.1356
D-009	0.1509	0.1709	0.1628
D-010	0.4845	0.4846	0.4608
D-011	0.3521	0.3847	0.3113
D-012	0.1979	0.2336	0.1827
D-013	0.0507	0.0701	0.0535
D-014	0.1528	0.1625	0.1476
D-015	0.3232	0.3389	0.3121
D-016	0.1964	0.2194	0.2012
D-017	0.2373	0.2690	0.2303
D-018	0.1742	0.1825	0.1743
D-019	0.3080	0.3541	0.3233
D-020	0.2888	0.2961	0.2844
D-021	0.1452	0.1601	0.1379
D-022	0.1999	0.2259	0.1839
D-023	0.2587	0.2741	0.2496
K-001	0.4072	0.4087	0.3984
K-002	0.2332	0.2610	0.2672
K-003	0.6277	0.6784	0.6706
K-004	0.4209	0.4455	0.4882
K-005	0.5746	0.5966	0.6365
K-006	0.5449	0.5599	0.5650
K-007	0.1971	0.2198	0.2213
K-008	0.3176	0.3505	0.2997
Average	0.2743	0.2930	0.2765

(Vaessen, 2018) in Python. This *WER* score reflects the amount of deviation between the transcription and the original prompt on a scale from 0 to 1. A higher number represents a greater error rate and more deviation. Participants with lower scores in Table 1 performed better (D-013 had the best performance, K-003 the worst). As mentioned earlier, any response that received a *WER* greater than 1 was truncated to 1 so that there could not be >100 % error. We then calculated the intraclass correlation coefficient (ICC) between the two human raters and then again including Whisper ASR.

We have made our entire data set as well as user friendly versions of our scripts available on our OSF project page (<https://osf.io/837pk/>), though you will need to provide your own Replicate API token to use Whisper.

RQ2: EI Test WER Scoring Procedure

Our second research question aimed to validate *WER* as a scoring metric for EI responses by comparing it to the traditional method of manual scoring using the Ortega ordinal scale rubric. The Ortega scoring system is accepted practice within the field and has been validated in a number of studies, so we use it as a benchmark to compare against.

Both authors scored all 900 EI response recordings using the 5-point Ortega scale while referring to their own transcriptions. ICC between the two raters was 0.808, 95 % CI [0.784, 0.830] showing strong but not perfect agreement. For any sentences which both raters had heard identically, differences were resolved through discussion. Differences in scoring due to different understandings of what the participant had said were left alone. Each participant's overall EI test score was obtained by averaging all 30 of their response scores for both *WER* and Ortega methods (both raters' scores were averaged together). We then tested for correlation between the average Whisper *WER* scores and human Ortega scores for each participant, as well as the correlation between Whisper *WER* scores and human Ortega scores across all 900 items.

Results

RQ1: Whisper ASR Validation

RQ1 aimed to determine if Whisper ASR transcription can transcribe Japanese L1 English L2 learner EI responses as well as humans. Table 1 shows the average *WER* of each transcriber across the 30 item EI test for each participant. This data is also visualized in Fig. 4, sorted from lowest to highest *WER*. Participants with a D at the beginning of their ID were tested at the first author's university and those with K at the second author's university.

We calculated the ICC of *WER* scores across all 900 responses as well as across overall participant scores, summarized in Table 2. Across all items, ICC = 0.951, 95 % CI [0.944, 0.957] between the two human raters, indicating very strong agreement in the amount of error, despite word-level differences in rater transcriptions. The ICC between the two human raters and Whisper was slightly lower at ICC = 0.929, 95 % CI [0.921, 0.936] but still very strong. When calculated across overall participant scores, ICC was even stronger at ICC = 0.998, 95 % CI [0.995, 0.999] between the two raters and ICC = 0.997, 95 % CI [0.994, 0.998] when Whisper ASR is included. This near perfect ICC strongly suggests that Whisper ASR can transcribe EI item responses as well as human raters.

As can be seen by examining scores between raters in Table 1, Whisper sometimes results in slightly smaller error rates. As will be

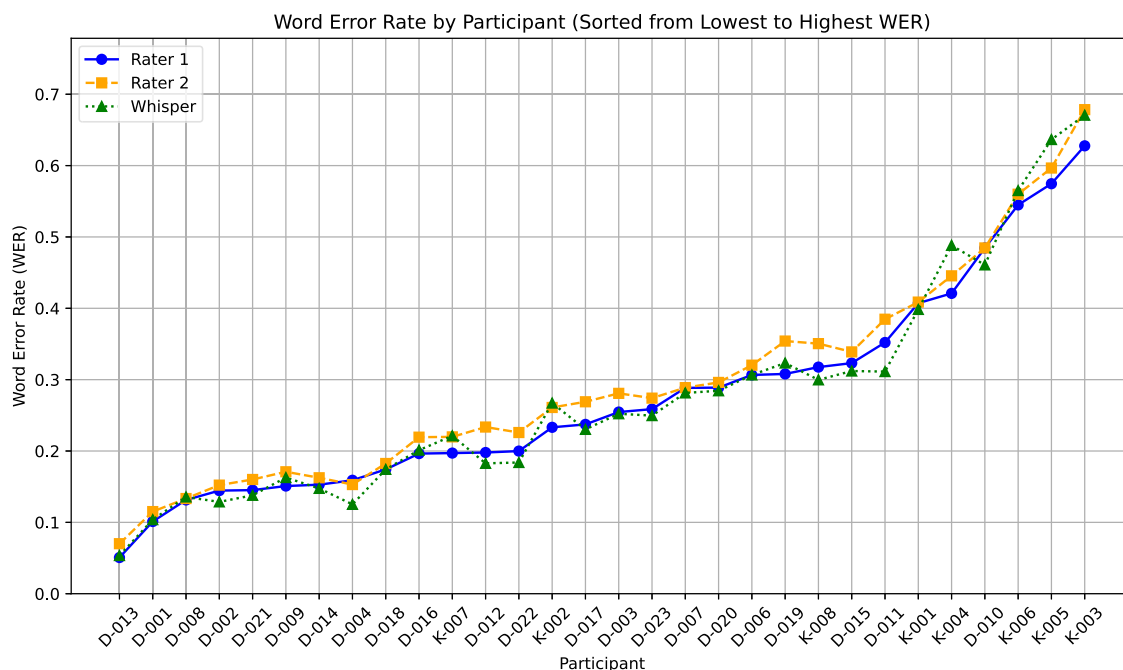


Fig. 4. Average *WER* for each transcriber for 30 participants sorted from lowest to highest average *WER*. Higher *WER* scores are equivalent to lower proficiency.

Table 2

Intraclass Correlation Coefficients between raters across all items and average.

Comparison	ICC	95 % CI
ICC for All 900 Items		
Between Rater 1 and Rater2	0.951	[0.944, 0.957]
Between Rater 1, Rater 2, and Whisper ASR	0.929	[0.921, 0.936]
ICC for 30 Overall Participant Scores		
Between Rater 1 and Rater2	0.998	(0.995, 0.999)
Between Rater 1, Rater 2, and Whisper ASR	0.997	(0.994, 0.998)

shown below, this is likely because Whisper attempts to reduce grammatical infelicities in its transcription and select probable words with no reference to the stimulus.

Looking at differences between the transcribed sentences for participants whose productions were clear, there were not large differences in the human raters and Whisper. For example, for participant D-001, there were occasionally small discrepancies like this:

Prompt: the employee was hired at a different nearby location

Rater 1: the employee was hired by at a different [X] location

Rater 2: the employee was a hired by a at a different prior location

Whisper: the employee was hired at a different party location

It is impossible at this point to say who is exactly correct in their transcription. Rater 1 could not understand the word between ‘different’ and ‘location’ while Rater 2 thought that ‘prior’ was uttered. Whisper interpreted it as ‘party’. However, all agree there was some deviation away from the prompt.

Consistent and accurate rating is more difficult with participants who scored lower overall. Participant K-003 performed the worst and was often unintelligible. Because of this, there were significant differences between all the transcribers, whether human or not, as seen in Table 3. In the case of the fourth prompt in Table 3, even the human raters can hear quite different productions when they are not forcing the data to conform to the original prompt. While there can be some large differences in *WER* between raters on individual responses, their averages across the entire 30-item test are very close, as seen in Table 1.

Whisper has a built-in normalization algorithm which sometimes omits repetitions and revisions such as deleting the ‘people’ repeated twice in the third prompt in Table 3, but the amount of deviation away from the original prompt is still captured. It may also try to conform to grammatical rules when transcribing unintelligible responses, such as in the third prompt. This normalization can be disabled but it results in much more irregular (sometimes bizarre) transcriptions. You can also provide Whisper with an initial prompt for forced alignment, but the results do not change significantly. After trying many variations, we found that the default settings performed the most reliably.

RQ2: WER Validation

For RQ2, we examined the correlation between EI test scores using the two different scoring methods: *WER* scoring of Whisper ASR transcriptions and traditional Ortega scoring done by human raters. Because manual Ortega scoring has been accepted as standard practice, we are using it as a benchmark to validate *WER* scoring against. Because *WER* is an error rate, it has a negative correlation with Ortega scoring. Given the results from RQ1 which suggested that Whisper ASR can transcribe as accurately as humans, if the Whisper ASR *WER* scoring method correlates highly with manual Ortega scoring, it points to the feasibility of automated EI testing using the Whisper ASR and *WER* scoring.

Correlation between Whisper *WER* scores and human Ortega scores was tested across all items as well as across overall participant scores. The correlation across all items was strong at $r = -0.817$, 95 % CI $[-0.838, -0.794]$, and it was even more pronounced across overall participant scores (using the mean score for each of the participants) at $r = -0.969$, 95 % CI $[-0.985, -0.935]$. The lower correlation at the item level is due to differences in the item-level granularity of the two scoring methods. While *WER* calculates a score for each item anywhere between 0 and 1, Ortega scoring only has five options with a sizable gap between them. Ortega scoring only achieves granularity when averaging the scores across the entire 30 item test. So, while there are differences at the item level, the outcome of *WER* scoring and Ortega scoring are highly similar when evaluating overall test scores, which is the primary concern in most assessment contexts. The correlation is visualized in Fig. 5 below.

Table 3

Comparison of transcriptions and their WER for participant K-003. Higher WER scores indicate more deviance away from the prompt.

Prompt	Rater 1 Transcription	WER	Rater 2 Transcription	WER	Whisper Transcription	WER
some illnesses are quite unusual	some [X] usual	0.8	some kids usual	0.8	so cute	1
there are a number of churches in the town	there are number of churches in town	0.22	there are a number of churches in town	0.11	journal number of churches in town	0.44
people know him as a very independent and strong man	people strong man	0.7	people drawn one	0.9	people are becoming stronger	0.9
some experts think that this may be where wine was produced for the first time	wine produce this time	0.87	why employers [X] this time	0.87	I'm produced this time	0.87

Discussion

This study intended to test the feasibility of scoring an English elicited imitation (EI) test without human raters by using the Whisper automated speech recognition (ASR) system to transcribe the responses of English L2 learners and then subsequently use the Word Error Rate (WER) computational metric to describe the amount of deviation from the ‘ground truth’ of the original prompt sentence. We have explained that the ASR accuracy required to carry out such an analysis has, until now, proved inadequate, but is in fact now available and working quite well. Better yet, as open-source software, Whisper ASR can be used at little to no cost. The results of our first research question found that Whisper ASR approaches human-level performance in transcribing EI responses, with ICC = 0.929, 95 % CI [0.921, 0.936] between Whisper and two raters across all items. However, Whisper transcriptions of speakers with low language proficiency were less accurate than those of speakers with higher proficiency. This is not unexpected, as human raters also encountered difficulties transcribing these same responses. Speech from low-proficiency speakers presents inherent challenges for transcription that may affect WER measurements regardless of whether the transcriber is human or machine. Despite this, the results of our study suggest that ASR can perform at levels competitive with humans for EI transcription while offering greater reliability and consistency across multiple administrations—a significant advantage for large-scale language testing applications.

One concern that still needs to be addressed is more diverse audio recording quality. Our test was not conducted in a strict lab-based environment; all participants completed the test at the same time in the same room, and the audio capture was done through a website rather than directly to audio recording equipment. Background noise such as the voices of other participants and even a class bell was present in many of the recordings, yet Whisper ASR still transcribed as well as the human raters. If participants can take an EI test on their smartphones through a website, then the audio quality and amount of background noise may be less predictable. However, human raters would also struggle to transcribe low quality, unintelligible, or noisy recordings.

Another important note is that we have only looked at transcriptions of the English language. Some recent studies have tested the accuracy of Whisper ASR across many different languages (Amorese et al., 2023; Graham & Roll, 2024) and found some variability. 65 % of the training data used to develop Whisper was in English (Radford et al., 2022), so it is understandable that accuracy is the highest for native English speech. However, the results of our study show that it performs very well on the L2 English production of Japanese L1 speakers. Further testing of other languages in the EI context is needed.

We have also shown that using the WER metric on Whisper ASR transcriptions results in overall EI test scores that align very strongly ($r = -0.969$, 95 % CI [-0.985, -0.935]) with the accepted method of human raters manually scoring with the Ortega ordinal-scale rubric. This is very similar to the highest correlations reported by Isbell et al. (2023) of different combinations of ASR and scoring metrics with human Ortega scoring on a Korean EI test. It may be worthwhile comparing Whisper ASR with other ASR systems in combination with WER for EI scoring, but we have shown that Whisper ASR works very well and is quite accessible.

We are currently developing a web-based EI testing platform (<https://autoeit.org> - a sample test is available for researchers to try the interface). At the moment, the platform is only used for conducting EI tests and collecting recordings, but we will fully automate the

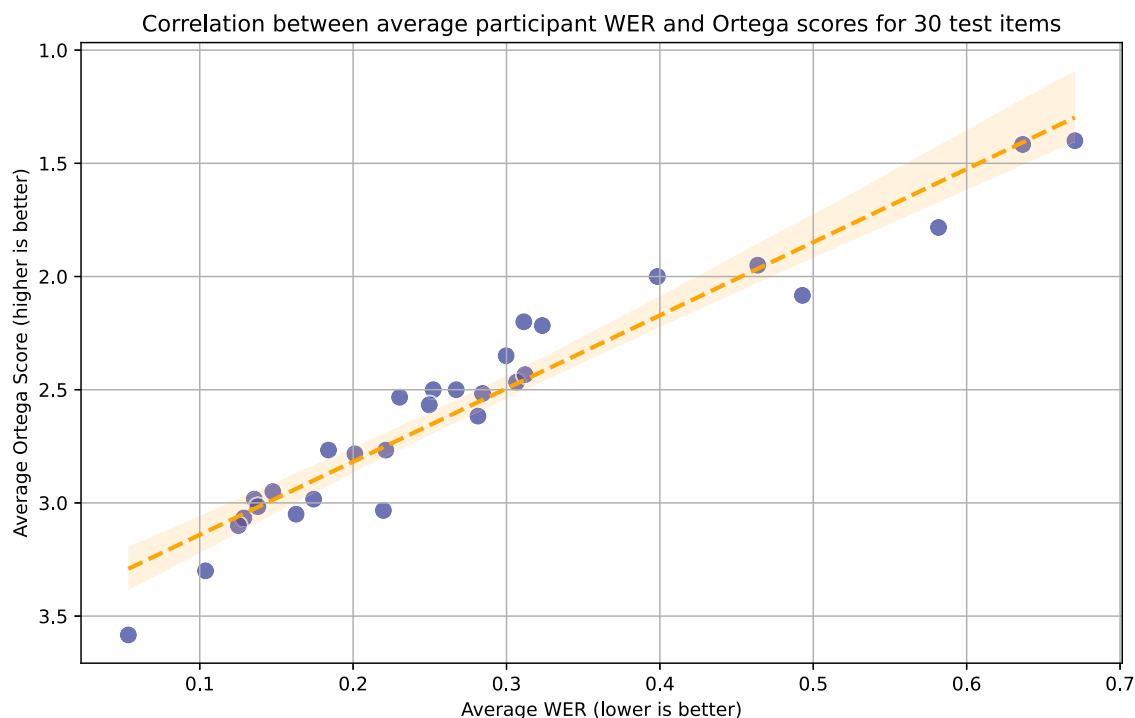


Fig. 5. Correlation between average participant WER and Ortega scores for 30 test items.

platform using Whisper ASR and WER so that tests can be taken on a smartphone and scoring can be done online in real time. We hope to make the fully automated platform available to other researchers and educators in the near future. We also plan to develop a much larger annotated item pool that we can use for English EI test form creation. We plan to use our automated platform to test a large number of sentences to determine relative difficulty and explore what linguistic features contribute to item difficulty in specific syllable bands. Having a large pool of valid and reliable sentences would allow for the creation of multiple EI test forms with equal difficulty which would make longitudinal testing more reliable. This would also let us create curriculum specific tests which focus on key vocabulary and grammatical features. With enough data, it would make conditions easier for creating EI tests that could be adaptive and pinpoint oral proficiency even more quickly than a standard EI test can. While we are currently focused on EI for English L2 learners, we hope to collaborate with other researchers who are interested in developing new EI tests in other languages which could be conducted on our automated platform.

CRedit authorship contribution statement

Michael McGuire: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jenifer Larson-Hall:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We would like to thank our participants for their contribution to this research. Funding for this research was provided by Doshisha University, Kyoto, Japan.

Appendix A Prompts used in the EI test

Prompt	Syllables
computers can calculate quickly	9
he never wrote any more music	9
some machines do not have parts that move	9
some illnesses are quite unusual	10
many books have been written about him	10
i cannot do my work without a car	10
a modern piano sounds quite different	11
both children and adults enjoy arts and crafts	11
there are a number of churches in the town	11
nuclear power plants are very expensive	12
fishing is an important business in this town	12
we do not know the answer to the question yet	12
the economy is mostly agricultural	13
every part of the body needs blood to survive	13
making chocolate is a process that has many steps	13
many people waited to enter the park that morning	14
he did not like school so he tried to avoid attending	14
she does not do anything except go to school and work	14
another economic activity is tourism	15
people know him as a very independent and strong man	15
if a house is near to the coast we say it is on the coast	15
heatstroke is a very serious medical emergency	16
most languages have ways to express differences of number	16
books have been written about what it was like living in the camp	16
the employee was hired at a different nearby location	17
some experts think that this may be where wine was produced for the first time	17
some things that are crimes in one country are not crimes in other countries	17
some animals look different even though they are closely related	18
today nobody knows what happened to the people who were living there	18
in a tea ceremony people drink tea out of bowls instead of cups	18

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